ANALYTICAL CUMULATIVE EXAMINATION
7th October 1995

The analysis of planetary objects that can only be viewed at great distances (meaning objects in orbit in the solar system, not just planets) poses particular challenges to the analytical chemist. However, several methods are been developed and some have been the subject of departmental seminars. In the last few decades some of the most rudimentary principles of the spectroscopy of inorganic solids have been applied to the determining the composition of asteroids.

1. (40%) Figure 1 shows the spectra of light reflected from the surface of four minerals, iron-nickel alloy, olivine ($(Fe,Mg)_2SiO_4$), pyroxene ($(Fe,Mg)SiO_3$) and spinel ($MgAl_2O_4$). Describe in the broadest terms (i) the origins of the features in these spectra and (ii) the overall shape of the curves.

2. (30%) Figure 2 shows a number of spectra obtained from the sunlight reflected from the surfaces of asteroids. Discuss these spectra using (i) the spectra shown in Fig. 1 (ii) the spectra for meteorites shown in Fig. 3.

3. (30%) Discuss any other methods that occur to you for determining the composition of planetary objects that cannot be sampled for laboratory analysis.

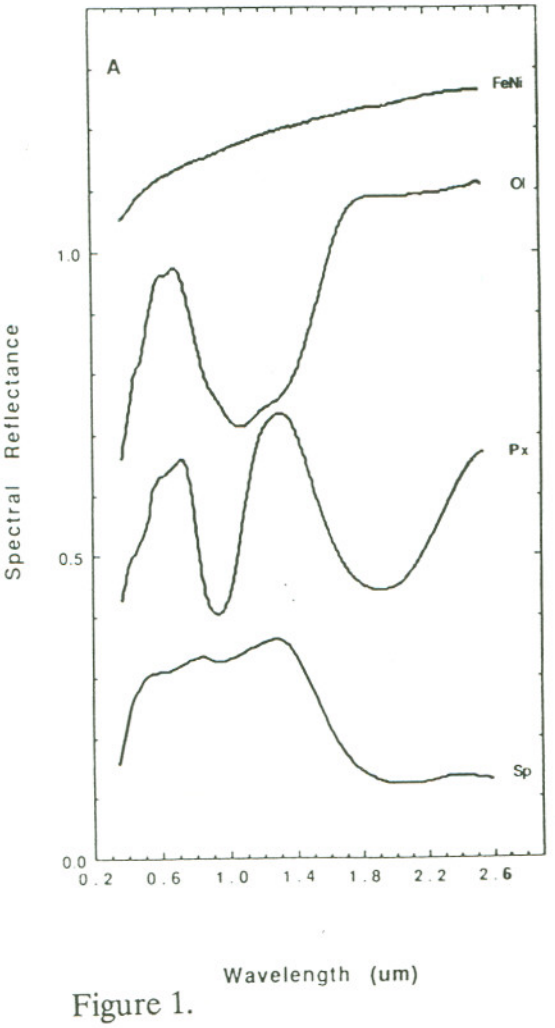

Figure 1.

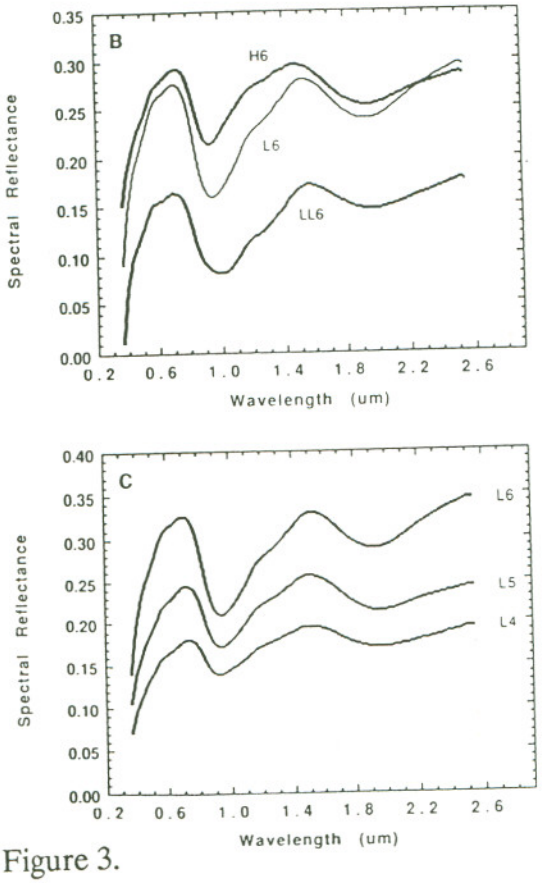

Figure 3.